

TABLE I: Performances of compared anonymization methods, including the EERs (%), WERs (%), and UARs (%) obtained in the ASV, ASR, and SER evaluations. EERs are presented for the development (dev) and test subsets of LibriSpeech (libri), and for male (m) and female (f) genders, respectively. The average EERs obtained across the evaluation subsets are presented in the row of *avg* for each method. The comparison between the baseline methods of random selection (RS), Average, PSD, and GAN-based methods, and the proposed IDMap-MLP, IDMap-Diff models is presented. Results obtained using both uniform ($\mathcal{U}$) and standard normal ($\mathcal{N}$) distributions for input vector sampling in the GAN-based methods, along with identity vector sampling in the proposed IDMap-MLP and IDMap-Diff models, are included.

	Dataset	Gender	RS	Average	PSD	GAN		IDMap-MLP		IDMap-Diff	
						$\mathcal{N}$	$\mathcal{U}$	$\mathcal{N}$	$\mathcal{U}$	$\mathcal{N}$	$\mathcal{U}$
EER	libri-dev	f	41.47	41.79	45.49	42.47	43.59	46.16	46.32	48.47	48.21
		m	43.22	39.60	44.31	44.22	40.77	46.60	44.24	47.86	47.36
	libri-test	f	41.54	39.43	43.43	43.54	44.62	43.61	43.81	48.18	47.66
		m	40.58	39.87	42.29	40.60	41.99	45.83	46.32	48.46	47.34
	avg		41.70	40.17	43.88	42.70	42.74	45.54	45.17	48.24	47.64
WER	libri-dev	-	3.31	3.37	3.39	3.38	3.45	3.37	3.38	3.38	3.41
	libri-test	-	3.21	3.23	3.23	3.28	3.23	3.23	3.25	3.22	3.28
SER	IEMOCAP-dev	-	53.47	52.25	53.42	53.48	52.74	53.67	52.79	53.78	54.23
	IEMOCAP-test	-	52.21	52.11	53.88	53.43	53.23	52.63	53.88	52.85	53.01